

Standardization (Z-score scaling)

Formula: $z = (x - \mu) / \sigma$

Let's say we have heights and weights of five people:

Person	Height (cm)	Weight (Kg)	Z Score (Standardized)
A	150	50	-1.26
B	160	60	-0.63
C	170	70	0
D	180	80	0.63
E	190	90	1.26

mean = 0
0 <= std <= 1

Min–Max Scaling (Rescaling to [0, 1])

Formula:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Let’s say we have heights and weight of five people:

Person	Height (cm)	Weight (Kg)(x)	Scaled (x')
A	150	50	0
B	160	60	0.25
C	170	70	0.5
D	180	80	0.75
E	190	90	1

50
0
20
40
.5
90-50
40
2 40
40
2 1

Robust Scaling (Outlier-Resistant)

Formula:

$$x' = \frac{x - \text{median}}{\text{IQR}}$$

where $\text{IQR} = Q3 - Q1$

Let's say we have heights and weight of five people:

Person	Height (cm)	Weight (Kg)(x)	Scaled (x')
A	150	50	-1
B	160	60	-0.5
C	170	70	0
D	180	80	0.5
E	300	200	6.5

IQR -->

Manually: Tukey

Q1: $(50+60) / 2 = 55$

Q3: $(80+200) / 2 = 144$

IQR = $140 - 55 = 85$

Pandas: Interpolation

Q1: 60

Q3: 80

--> uses plot values

Nominal vs Ordinal + One-Hot Encoding

Nominal: Categories with no order

Ordinal: Clear ranking

Tiny dataset

id	color	size	price
1	red	Small	10
2	blue	Medium	12
3	green	Large	15
4	red	Medium	11

One-hot the nominal feature by hand

Step 1: Creating New columns:

- Color_red, color_blue, color_green

Step 2: The transformed table:

id	price	color	Color_red	Color_blue	Color_green
1	10	red	1	0	0
2	12	blue	0	1	0
3	15	green	0	0	1
4	11	red	1	0	0

Step 3: Row by row visualization:

- id 1 red $\rightarrow [1,0,0]$
- id 2 blue $\rightarrow [0,1,0]$
- id 3 green $\rightarrow [0,0,1]$
- id 4 red $\rightarrow [1,0,0]$

Ordinal Encoding

Ordinal: Clear ranking

Tiny dataset

id	color	size	price
1	red	Small	10
2	blue	Medium	12
3	green	Large	15
4	red	Medium	11

Ordinal encoding for ordinal feature: size

Step 1: Decide Order:

Small: 1, Medium: 2, Large: 3

Step 2: The transformed table:

id	color	size	price	Size_encode
1	red	Small	10	1
2	blue	Medium	12	2
3	green	Large	15	3
4	red	Medium	11	2

Euclidean distance & Manhattan distance

What these measure

- **Euclidean distance:** straight line closeness.
- **Manhattan distance:** city-block closeness.

Formulas

For $p = [p_1, p_2, \dots, p_k]$, $q = [q_1, q_2, \dots, q_k]$

- Euclidean: $d_2(p, q) = \sqrt{\sum (p_i - q_i)^2}$
- Manhattan: $d_1(p, q) = \sum |p_i - q_i|$

Tiny dataset

$S_1 = [70, 80]$

$S_2 = [60, 90]$

$S_3 = [85, 60]$

$S_4 = [78, 76]$

$S_5 = [62, 65]$

$q = [75, 70]$ # query point

S_1 vs q

- Diffs: $(70 - 75) = -5$, $(80 - 70) = 10$
- Euclidean: $\sqrt{25 + 100} = \sqrt{125} \approx 11.180$
- Manhattan: $|-5| + |10| = 5 + 10 = 15$